Voice-Based Parkinson's Disease Classification Using Handcrafted Acoustic Features and Log-Mel Spectrograms

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***Abstract*—Parkinson's disease (PD) can affect speech production, motivating research into voice recordings as a non-invasive source of acoustic information. This study compares a classical machine-learning pipeline based on handcrafted acoustic summaries with a convolutional neural network (CNN) based on Log-Mel spectrograms. The experiments use an Italian Parkinson's Voice and Speech collection containing 831 WAV recordings from 65 participants: 37 healthy controls and 28 participants with PD. Participants, rather than individual recordings, are assigned to disjoint development and test partitions. Four classical classifiers are evaluated using five-fold participant-stratified cross-validation, and an RBF-kernel support vector machine (SVM) is selected. A separate CNN processes five-second, 64-band Log-Mel inputs. On 226 held-out recordings from 17 participants, the SVM achieves 0.9558 recording-level accuracy and 0.9412 participant-level accuracy; the CNN achieves 0.9690 and 0.9412, respectively. The SVM has a higher recording-level ROC-AUC, while both models have identical participant-level metrics in the saved run. The small held-out cohort and absence of external validation limit generalisation. This work is an experimental research prototype, not a clinical diagnostic tool.**

***Index Terms*—Parkinson's disease, voice analysis, acoustic features, support vector machine, convolutional neural network, Log-Mel spectrogram, participant-level evaluation.**

I. Introduction

Parkinson's disease (PD) is a progressive neurological disorder that can affect speech and voice production. Changes in phonation, articulation, timing, and prosody motivate research into acoustic measurements as potential indicators of motor and speech impairment. Voice analysis is attractive as a non-invasive and comparatively accessible research modality; however, a computational classifier trained on a limited dataset should not be interpreted as a clinical diagnostic instrument.

A central methodological issue is the repeated-measures structure of speech datasets. Multiple recordings from one person are correlated and may share speaker-specific characteristics. If recordings from the same participant are divided between training and testing, a model may exploit person-specific cues and produce an optimistic estimate of performance on unseen people. The present study therefore separates participants before model development and reports both recording-level and participant-level metrics.

The research question is: how do a classical machine-learning pipeline based on statistical summaries of handcrafted acoustic features and a convolutional neural network (CNN) based on Log-Mel spectrograms compare when tested on participants excluded from model development? The objectives are to audit the available cohort, implement both representations, compare classical classifiers using participant-grouped cross-validation, and evaluate the selected models on the same held-out participants.

The contribution is an implementation-level comparison on the Italian Parkinson's Voice and Speech collection using participant-disjoint evaluation, dual-level reporting, and exploratory feature-importance analysis. The study does not claim a new clinical biomarker or superiority over published methods.

II. Related Work

Research on voice-based PD classification has followed two broad directions: handcrafted acoustic descriptors paired with conventional classifiers, and learned representations processed by neural networks. Reviews by Ngo *et al.* and Amato *et al.* describe a diverse literature spanning speech tasks, signal descriptors, and statistical or machine-learning classifiers [2], [3]. These reviews also make clear that studies differ in their datasets, recording protocols, and evaluation procedures, limiting direct comparison of headline accuracy values.

Handcrafted pipelines commonly summarise measurable properties of the voice signal and use classifiers such as support vector machines (SVMs), random forests, or neural networks. Their advantage is that the input variables have interpretable acoustic definitions and can be inspected using feature analysis. Their limitation is that the selected descriptors may not preserve all temporal and contextual information in a speech signal. Amato *et al.* survey feature-based machine-learning methods and discuss the role of acoustic features in PD voice analysis [3]. In the present work, MFCC, delta-MFCC, spectral, energy, and chroma summaries form the tabular representation, with SVM and tree-based classifiers used as comparators.

A second direction uses neural networks to learn discriminative patterns from structured signal representations. A Log-Mel spectrogram retains local time-frequency structure, allowing convolutional filters to learn patterns across neighbouring time and frequency regions. Such learned representations are flexible, but their performance depends on data volume, architecture, recording conditions, and the evaluation protocol. The

current study uses a compact CNN as a controlled alternative to the handcrafted-feature pipeline rather than claiming that deep learning is inherently superior.

Luna-Ortiz *et al.* proposed an improved smallest normalized difference associative memory (ISNDAM) method, incorporating a feature-relevance identification phase, and compared it with a large set of classifiers on two established voice datasets [1]. Their paper is the supplied base reference for this project. The method in the present notebook is different: it implements Logistic Regression, RBF-SVM, Random Forest, Extra Trees, and a Log-Mel CNN. ISNDAM is not implemented here, and the published ISNDAM results are not results of this project.

Recent reviews reinforce the need to interpret high reported performance carefully. Ngo *et al.* reviewed computerized speech and voice analysis and highlighted the variation in datasets and analysis methods [2]. Xavier *et al.* reviewed voice analysis studies published between 2019 and 2023, showing the breadth of approaches and tasks in the field [4]. A 2025 systematic review by Sedigh Malekroodi *et al.* similarly describes a field with many reported high-performing models but persistent concerns around dataset heterogeneity, validation, calibration, and clinical translation [5]. These reviews motivate transparent reporting of the participant split and the unit of evaluation.

The dataset used here is the Italian Parkinson's Voice and Speech (IPVS) collection. Hossain, Traini, and Amenta describe the IPVS dataset as comprising 65 native Italian speakers divided into 15 young healthy controls, 22 elderly healthy controls, and 28 participants with PD [6]. This matches the cohort counts found in the project notebook. Their work is particularly relevant because it concerns the same named dataset; however, its reported methods and outcomes must remain distinct from the experiments in this paper.

A. Comparative Synthesis of Prior Work

The selected literature provides complementary rather than directly interchangeable evidence. The base study by Luna-Ortiz *et al.* investigates associative-memory-based classification and feature relevance on voice datasets; it motivates the present project but does not describe the models evaluated here [1]. The systematic review by Ngo *et al.* places computerized speech analysis in a broad methodological context and emphasizes the diversity of voice tasks and signal descriptors used in the field [2]. Amato *et al.* focus specifically on machine-learning and statistical approaches, making their survey relevant to the handcrafted-feature branch of this work [3].

More recent reviews broaden the comparison to deep-learning representations and validation practices. Xavier *et al.* review studies published from 2019 to 2023 and document the range of voice-analysis methods used for diagnosis and prognosis [4]. Sedigh Malekroodi *et al.* synthesize 69 studies published between 2020 and September 2025, grouping evidence by dataset, feature type, model family, and validation strategy [5]. Their review describes the coexistence of classical machine learning, deep learning, and hybrid methods, while identifying dataset heterogeneity and inconsistent validation as recurring obstacles. These observations support the decision in the present study to compare two representation families under a common participant-disjoint test partition.

The IPVS-specific study by Hossain, Traini, and Amenta is especially relevant for understanding the cohort composition and recording collection [6]. However, its experimental pipeline is not the same as the notebook used here. In particular, the present work reports its own preprocessing, feature construction, classifiers, split, and test metrics. Published performance values from that paper are therefore not copied into the results of this project.

B. Research Gap and Positioning

The literature does not support claiming that voice-based PD classification is unexplored. Rather, the practical question addressed here is narrower: within the available IPVS project data, how do a compact handcrafted-feature pipeline and a compact Log-Mel CNN compare when participants are kept disjoint between model development and final testing? The paper's value lies in documenting this implementation and reporting both file-level and participant-level outcomes. A fair comparison with other publications would require matching cohorts, speech tasks, preprocessing, participant partitions, and evaluation units; such a matched cross-study experiment is outside the scope of this project.

III. Dataset Description

The notebook audits three cohorts: young healthy controls (YHC), elderly healthy controls (EHC), and participants with PD. The corrected audit identifies 65 unique participants and 831 WAV recordings. The two healthy-control cohorts are combined into the binary healthy-control (HC) class. At recording level, there are 394 HC and 437 PD recordings; at participant level, there are 37 HC and 28 PD participants.

The recording counts are not evenly distributed across participants. In particular, the elderly healthy-control and PD cohorts contribute substantially more recordings than the young healthy-control cohort. Consequently, a recording-level metric weights participants with many recordings more heavily unless an explicit participant aggregation or weighting scheme is used. The notebook addresses this issue in two ways: participants are kept intact during partitioning, and the final report includes participant-level metrics in addition to recording-level metrics. The distinction is important because the number of recordings is not the same as the number of independent people.

The two healthy-control cohorts are combined for the binary classification task, but their original cohort identities are retained during the audit. This preserves the ability to describe the age-cohort composition and highlights a potential source of confounding. The available experiment does not establish that the model separates disease-related speech changes from age-related differences. This issue is treated as a limitation rather than being resolved by an unsupported subgroup claim.

TABLE I
QUALITATIVE SYNTHESIS OF SELECTED LITERATURE AND ITS RELATIONSHIP TO THE PRESENT STUDY

Study	Scope	Relevance to this project	Boundary of comparison
Luna-Ortiz *et al.* [1]	Associative-memory approach and feature relevance for voice-based PD classification	Supplied base reference; establishes prior work beyond standard classifiers	ISNDAM is not implemented in the project notebook; its results are not project results
Ngo *et al.* [2]	Systematic review of computerized speech and voice analysis	Provides context on signal descriptors, tasks, and evaluation diversity	Review-level synthesis, not a matched experimental baseline
Amato *et al.* [3]	Survey of machine-learning and statistical voice analysis	Supports discussion of handcrafted features and conventional classifiers	Surveyed datasets and protocols differ from the current split
Xavier *et al.* [4]	Systematic review of voice analysis studies from 2019–2023	Broadens the discussion to diagnostic and prognostic tasks	Different studies, cohorts, and metrics prevent direct score comparison
Sedigh Malekroodi *et al.* [5]	Systematic review of ML and DL approaches through September 2025	Informs comparison of handcrafted, spectrogram, and learned representations	Review findings are contextual evidence, not a benchmark for this experiment
Hossain *et al.* [6]	Machine-learning voice analysis using the IPVS collection	Relevant source for the dataset and cohort description	Their experimental pipeline and results are distinct from this notebook

TABLE II
PARTICIPANT AND RECORDING DISTRIBUTION FROM THE NOTEBOOK AUDIT

Cohort	Participants	Recordings
Young healthy controls	15	45
Elderly healthy controls	22	349
Parkinson's disease	28	437
Total	65	831

The directory scanner accounts for the different folder structures used by the cohorts. For the PD group, the range directory is retained in the participant identifier to avoid collisions between repeated folder names. Participant identifiers are used for grouping and auditing only; they are not supplied to the classifiers as predictive variables. The saved audit reports no unknown task codes and no feature-extraction errors.

IV. METHODOLOGY

A. Overall Workflow and Participant Split

Fig. 1 summarises the implemented workflow. After cohort and participant auditing, the participant table is stratified into training and test participants. All recordings from a participant remain in the same partition. The saved split contains 48 training participants (27 HC and 21 PD) and 17 held-out test participants (10 HC and 7 PD), corresponding to 605 training recordings and 226 test recordings. The held-out participants are not used for model selection.

For classical models, five-fold stratified cross-validation is conducted using training participants only. Each fold is formed at participant level, and all recordings from the fold's participants are mapped to the corresponding training or validation subset. Balanced accuracy is used to compare classifiers. This prevents participant overlap between folds, although the reported fold metric is calculated over the recordings belonging to each fold.

B. Experimental Design and Leakage Control

The participant is the grouping unit throughout the development process. The split is created from a participant-level table, stratified by the binary label, and then expanded back to the corresponding recordings. This order is deliberate: splitting individual WAV files first would permit recordings from one speaker to appear in both training and testing sets. Such overlap can make evaluation optimistic because the model may learn stable speaker-specific characteristics rather than patterns that transfer to an unseen person.

The classical-model comparison uses only the 48 development participants. Five participant-stratified folds are formed within this development group, and balanced accuracy is used to rank the candidate models. The held-out 17 participants are not involved in model selection. After the RBF-SVM is selected, it is evaluated on the held-out recordings. The CNN uses a separate participant-level validation split within the same 48 development participants: 38 participants for training and 10 for validation. The same 17 held-out participants are retained for its final evaluation. This design allows a comparison on a common test cohort while keeping the CNN validation group separate from its training group.

The split sizes are summarized in Table III. The participant counts, rather than the recording counts alone, should be used when interpreting the reliability of the final estimates.

C. Audio Preprocessing and Handcrafted Features

Audio is loaded from WAV files and converted to mono by averaging channels when necessary. Signals are resampled to 16 kHz, mean-centred, and peak-normalised when the peak amplitude is nonzero. For a discrete recording $x[n]$, mean-centering is

$$x_c[n] = x[n] - \bar{x}, \tag{1}$$

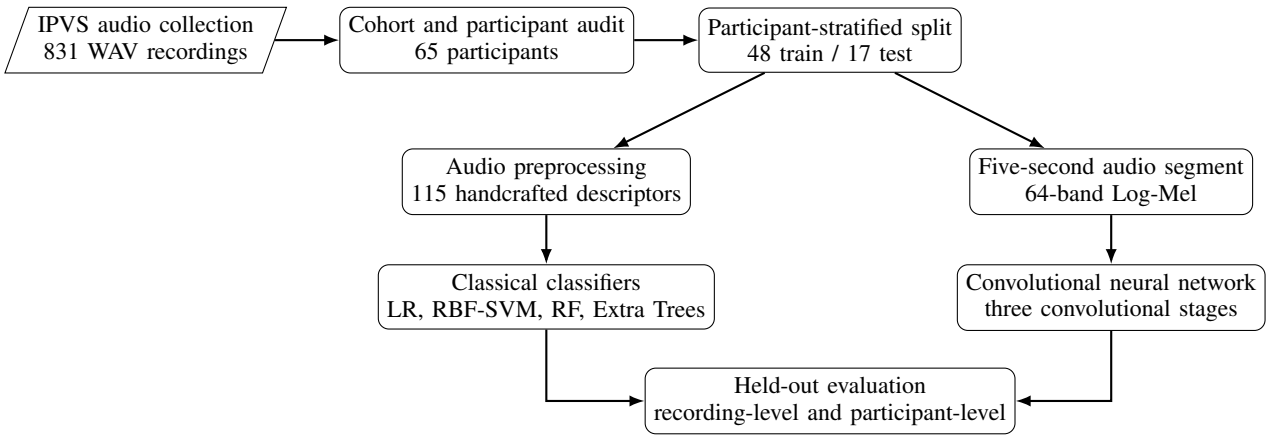


Fig. 1. Team-developed schematic of the two implemented audio-classification branches.

TABLE III
PARTICIPANT-DISJOINT PARTITIONS

Partition	Participants	Recordings
Development (classical models)	48	605
CNN training subset	38	–
CNN validation subset	10	–
Held-out test	17	226

and peak normalisation is

$$x_p[n] = \frac{x_c[n]}{\max_n |x_c[n]|}. \tag{2}$$

The normalisation scales the maximum absolute sample; it does not remove differences in microphone, room, or speaking conditions.

The notebook extracts 20 MFCC coefficients and their delta descriptors. It additionally computes spectral centroid, spectral bandwidth, spectral rolloff, zero-crossing rate (ZCR), root-mean-square (RMS) energy, and chroma features. For each frame-based feature sequence, the mean and standard deviation are calculated. Together with the duration feature, these operations produce 115 feature columns per recording.

MFCCs summarise the short-time spectral envelope. The signal is framed and windowed, transformed using the short-time Fourier transform, mapped through Mel-spaced filters, logarithmically compressed, and decorrelated using a discrete cosine transform. Delta-MFCCs describe temporal changes in the cepstral sequence. Spectral centroid estimates the weighted centre of spectral energy; bandwidth measures its spread; rolloff identifies a frequency below which a specified proportion of energy lies; ZCR counts sign changes; RMS summarises frame energy; and chroma maps spectral energy into pitch-class bins.

For a frame-level sequence $z_1, \ldots, z_T$, the summary statistics are

$$\mu_z = \frac{1}{T}\sum_{t=1}^{T} z_t, \qquad \sigma_z = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(z_t - \mu_z)^2}. \tag{3}$$

TABLE IV
HANDCRAFTED FEATURE FAMILIES USED BY THE CLASSICAL PIPELINE

Feature family	Acoustic information represented
MFCC and delta-MFCC	Spectral-envelope shape and temporal change in cepstral coefficients
Spectral centroid	Weighted centre of spectral energy
Spectral bandwidth	Spread of energy around the spectral centroid
Spectral rolloff	Frequency below which the selected proportion of spectral energy lies
Zero-crossing rate	Rate of sign changes in the waveform
RMS energy	Short-time signal energy
Chroma	Distribution of spectral energy across pitch-class bins
Duration	Recording length as a temporal summary

The fixed-length summaries permit conventional classifiers to process recordings with different durations.

Table IV summarizes the feature families and their roles. The notebook stores the frame-level descriptors as statistical summaries, producing a fixed-width vector for each recording. The feature vector is intended to represent complementary aspects of the signal: spectral envelope, short-term spectral change, energy, pitch-class distribution, and signal duration.

D. Classical Machine-Learning Models

Four classifiers are evaluated: Logistic Regression, an RBF-kernel SVM, Random Forest, and Extra Trees. Median imputation and standardisation are included in the Logistic Regression and SVM pipelines. The tree-based pipelines use median imputation without standardisation. Class weighting is enabled. The SVM is selected because it obtains the highest mean balanced accuracy in the five training-participant folds.

E. Log-Mel Spectrogram and CNN

For the CNN branch, each recording is resampled to 16 kHz and represented by a five-second segment taken from the beginning of the recording. Shorter signals are zero-padded and longer signals are truncated. The Mel spectrogram uses 64 Mel bands, an FFT size of 1024, and a hop length of 256

TABLE V
FIVE-FOLD PARTICIPANT-STRATIFIED CROSS-VALIDATION RESULTS

Classifier	Mean BA	Std. dev.
RBF-SVM	0.9606	0.0360
Extra Trees	0.9483	0.0474
Random Forest	0.9404	0.0469
Logistic Regression	0.9328	0.0302

TABLE VI
CNN CONFIGURATION

Component	Configuration
Input	$64 \times 313 \times 1$
Convolution filters	16, 32, 64
Kernel size	3×3
Pooling	Max pooling after stages 1 and 2
Aggregation	Global average pooling
Dense layer	32 units, ReLU
Dropout	0.45 and 0.35
Output	1 unit, sigmoid
Optimizer	Adam, 3×10^{-4}
Loss	Binary cross-entropy
Batch / maximum epochs	16 / 60

samples. Power values are converted to decibels relative to the maximum value. Each input has shape $64 \times 313 \times 1$.

The CNN contains three convolutional stages with 16, 32, and 64 filters. The first two stages are followed by batch normalisation and max pooling. The third stage is followed by global average pooling. The classification head contains dropout (0.45), a 32-unit ReLU dense layer, dropout (0.35), and a one-unit sigmoid output. The model uses Adam with learning rate 3×10^{-4} and binary cross-entropy. Training is limited to 60 epochs with batch size 16, early stopping (validation-loss patience 8), and learning-rate reduction on plateau (factor 0.5, patience 3, minimum learning rate 10^{-6}).

The 48 development participants are split into 38 training and 10 validation participants. The 17 held-out test participants remain unchanged. Sample weights are constructed so that participants contribute equally overall, followed by participant-level class balancing.

F. Evaluation Metrics and Participant Aggregation

The study reports accuracy, balanced accuracy, PD precision, PD recall, F1-score, and ROC-AUC. Accuracy is the fraction of correct predictions. Balanced accuracy is the mean of sensitivity and specificity. Precision measures the proportion of positive predictions that are correct; recall measures the proportion of PD cases detected; F1 is their harmonic mean; and ROC-AUC measures ranking performance across thresholds.

For participant-level evaluation, recording-level PD probabilities are averaged within each participant. If participant q has R_q recordings, the aggregate score is

$$\bar{p}_q = \frac{1}{R_q} \sum_{r=1}^{R_q} p_{qr}. \tag{4}$$

A participant is assigned to the PD class when $\bar{p}_q \geq 0.5$. This is a distinct evaluation unit from an individual recording.

V. EXPERIMENTAL RESULTS

The selected RBF-SVM is evaluated on 226 held-out recordings from 17 participants. The CNN is evaluated on the same test recordings. Table VII reports the metrics at both recording and participant levels.

At recording level, the CNN has higher accuracy (0.9690 versus 0.9558), balanced accuracy (0.9692 versus 0.9568), precision, and F1-score. Both models have the same PD recall (0.9714). The SVM has a higher ROC-AUC (0.9939 versus 0.9892). Thus, the comparison does not yield a single model that dominates every metric. No statistical significance test is implemented, so these numerical differences should not be described as statistically significant.

At participant level, both models have the same reported metrics: 0.9412 accuracy, 0.9500 balanced accuracy, 0.8750 PD precision, 1.0000 PD recall, 0.9333 F1-score, and 1.0000 ROC-AUC. The participant test set contains only 17 people, so one participant corresponds to approximately 5.9 percentage points of accuracy. The 226 recordings are repeated observations from those 17 participants and must not be interpreted as 226 independent subjects.

A. Confusion-Matrix Analysis

The recording-level confusion matrices in Fig. 2 provide the class-wise counts underlying the aggregate metrics. For the RBF-SVM, 114 healthy-control recordings are correctly classified as HC and 102 PD recordings are correctly classified as PD; seven HC recordings are predicted as PD and three PD recordings are predicted as HC. For the CNN, 117 HC recordings and 102 PD recordings are correctly classified, while four HC recordings and three PD recordings are misclassified. Thus, both models identify the same number of PD recordings in this test set, while the CNN makes fewer false-positive HC predictions. These counts describe recordings, not independent participants.

B. Task-Wise Descriptive Evaluation

The notebook also reports task-wise CNN results for the held-out recordings. These results are descriptive and should not be interpreted as independent task-specific validation: many task subsets contain only 13–17 recordings and participants. Table VIII reproduces the saved task-wise balanced accuracy and PD F1 values. The variability across tasks suggests that the acoustic content of the recording task may influence model performance. However, the small per-task counts prevent strong conclusions about which task is intrinsically easier or more clinically informative.

C. Exploratory Feature Importance

Permutation importance is computed for the selected SVM using balanced accuracy on the held-out test recordings. The largest mean importance values in the saved output include MFCC means, chroma variability, MFCC standard deviations,

TABLE VII
HELD-OUT TEST PERFORMANCE FOR THE TWO IMPLEMENTED MODELS

Model	Evaluation unit	n	Accuracy	Balanced acc.	PD precision	PD recall	PD F1 / AUC
RBF-SVM	Recording	226	0.9558	0.9568	0.9358	0.9714	0.9533 / 0.9939
CNN	Recording	226	0.9690	0.9692	0.9623	0.9714	0.9668 / 0.9892
RBF-SVM	Participant	17	0.9412	0.9500	0.8750	1.0000	0.9333 / 1.0000
CNN	Participant	17	0.9412	0.9500	0.8750	1.0000	0.9333 / 1.0000

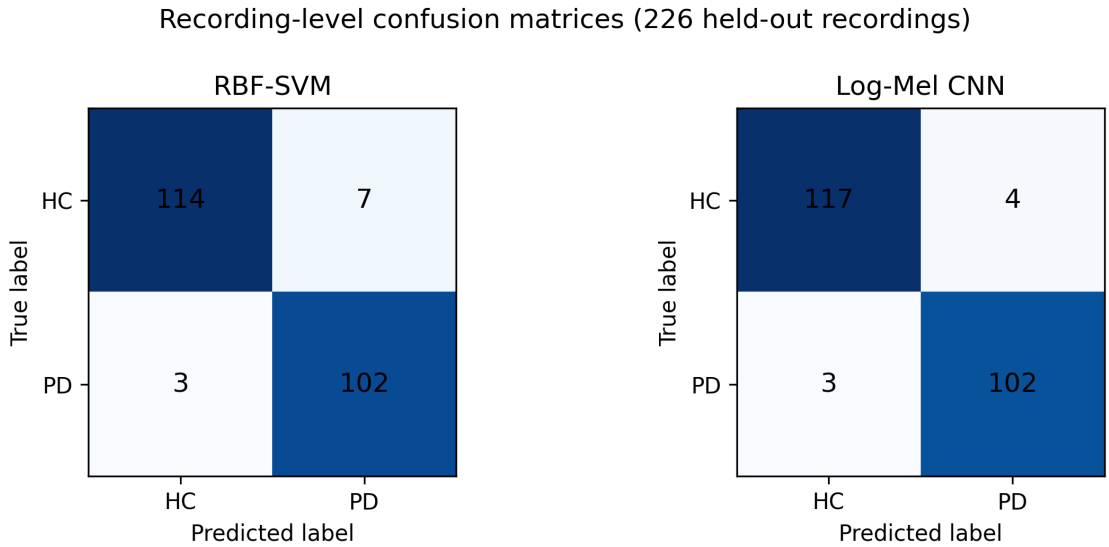


Fig. 2. Recording-level confusion matrices reconstructed from the notebook's held-out classification reports. Rows indicate true classes and columns indicate predicted classes.

TABLE VIII
TASK-WISE CNN RESULTS ON THE HELD-OUT SET (DESCRIPTIVE)

Task	Recordings	Balanced acc.	PD F1
B1	17	0.9500	0.9333
B2	15	0.9500	0.9091
D1	14	1.0000	1.0000
D2	14	1.0000	1.0000
FB1	14	1.0000	1.0000
PR1	17	0.9500	0.9333
VA1	14	0.9286	0.9231
VA2	13	1.0000	1.0000
VE1	14	1.0000	1.0000
VE2	13	0.8571	0.8571
VI1	14	0.9286	0.9231
VI2	13	0.8452	0.8333
VO1	14	1.0000	1.0000
VO2	13	1.0000	1.0000
VU1	14	1.0000	1.0000
VU2	13	0.9286	0.9231

delta-MFCC, spectral bandwidth, spectral rolloff, and RMS variability. Fig. 3 shows the ten highest-ranked entries in the saved output.

The ten highest-ranked features and their saved mean permutation-importance values are listed in Table IX. MFCC summary variables appear frequently among the leading features, alongside chroma variability and spectral bandwidth.

TABLE IX
TOP TEN SVM PERMUTATION-IMPORTANCE FEATURES

Feature	Mean importance
mfcc_10_mean	0.007178
mfcc_6_mean	0.006379
chroma_2_std	0.005301
mfcc_11_mean	0.004825
mfcc_1_mean	0.004789
mfcc_1_std	0.004762
mfcc_7_mean	0.004699
spectral_bandwidth_1_std	0.004412
delta_mfcc_11_mean	0.004258
mfcc_5_mean	0.003935

The reported values are changes in balanced accuracy under feature permutation, not coefficients, probabilities, or measures of biological effect. Several descriptors may be correlated, so the ranking should be read as model-specific exploratory evidence.

Permutation importance is conditional on the fitted model, dataset, and evaluation sample. Correlated descriptors may share predictive information. The ranking is exploratory and does not establish causality, a biological mechanism, or a clinically validated biomarker.

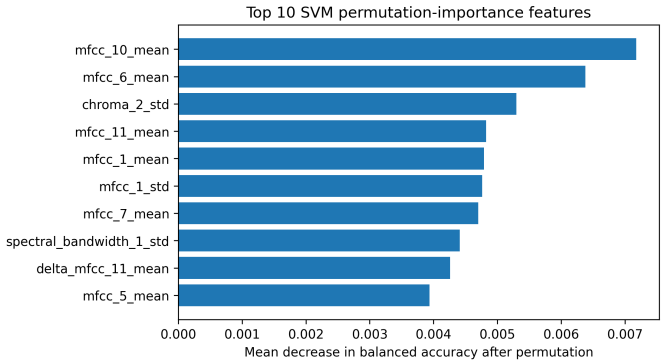


Fig. 3. Top ten SVM permutation-importance features from the saved notebook output.

VI. Discussion

The participant-aware split is an important design choice because it prevents recordings from the same individual from appearing in both training and held-out testing. The separate participant-level results provide a more direct view of performance on unseen people than file-level scores alone. Nevertheless, the participant test set is small, and its metrics have limited precision.

The two approaches show complementary behaviour. The CNN is stronger on several recording-level threshold metrics, while the SVM has the higher recording-level ROC-AUC. Their participant-level scores are identical in this saved run. These observations are specific to this dataset, split, and training procedure. They do not demonstrate that CNNs or SVMs are generally preferable for PD voice classification.

The study should also be interpreted in light of the related literature. Reviews describe substantial variation in datasets, language, speech tasks, feature definitions, and validation procedures [2]–[5]. Consequently, direct numerical comparisons with published results are not justified unless experimental conditions are matched. The ISNDAM paper is relevant background, but its datasets, algorithm, and statistical testing differ from the present implementation [1].

A. Practical Implications

The results support the feasibility of comparing compact acoustic-feature and spectrogram-based pipelines on a shared participant-disjoint evaluation set. The handcrafted pipeline has a smaller, inspectable feature representation and provides an exploratory ranking of input variables. The CNN learns its own local time-frequency filters but depends on a fixed five-second crop and a larger training procedure. Neither approach can be recommended as a clinical screening tool from this experiment alone. A practical next step is a larger, externally validated study with consistent recording instructions, demographic reporting, calibration analysis, and pre-specified participant-level endpoints.

B. Interpretation of Recording- and Participant-Level Results

The recording-level and participant-level metrics answer different questions. Recording-level evaluation measures how often an individual file is classified correctly, but participants contribute different numbers of files. Participant-level aggregation gives one prediction per person by averaging that person's recording probabilities. In this experiment, the participant-level accuracy is 0.9412 for both models, corresponding to 16 correct classifications among 17 held-out people. Because one participant changes accuracy by approximately 5.9 percentage points, small numerical differences should not be overinterpreted. The perfect participant-level ROC-AUC is also based on only 17 participants and should not be read as evidence of perfect generalization.

The task-wise descriptive results add another view of heterogeneity. Balanced accuracy ranges from 0.8452 to 1.0000 across the saved task subsets. Some tasks contain only 13 recordings, and these small subsets provide unstable estimates. The task analysis is therefore useful for identifying a question for future investigation—whether task type affects performance—but is not sufficient to establish a task ranking or to recommend a particular clinical protocol.

C. Reproducibility and Threats to Validity

Several implementation choices support reproducibility: a participant audit precedes modeling; participant identifiers are used for grouping rather than prediction; model selection is restricted to development participants; the held-out participants remain untouched until final evaluation; and the CNN uses a fixed input representation and recorded training configuration. The notebook also saves model artifacts, metrics, split information, and audit outputs. A future release should include the exact environment versions, random seeds, and a machine-readable participant split manifest so that the experiment can be independently reconstructed.

The principal threat to internal validity is the limited number of independent participants. Grouped splitting reduces leakage between speakers, but it does not eliminate uncertainty caused by a small test cohort or a single saved partition. The classical model comparison uses grouped folds, whereas the final test estimate is still a single split. A stronger follow-up would repeat participant-grouped outer splits or use nested grouped validation, reporting the distribution of participant-level metrics and confidence intervals.

Construct validity is also limited. The model predicts the dataset label from audio recordings; it does not directly measure disease severity, progression, treatment response, or clinical diagnosis. The healthy-control group combines young and elderly cohorts, and the experiment does not isolate age effects. The CNN uses only the first five seconds of each recording, which may not capture all relevant content in longer tasks. Finally, the IPVS collection consists of native Italian speakers and a defined recording protocol, so performance should not be assumed to transfer to other languages, microphones, or unsupervised home recordings.

TABLE X
KEY EXPERIMENTAL SETTINGS RECORDED IN THE NOTEBOOK

Setting	Implemented configuration
Audio sample rate	16 kHz
Classical input	115 handcrafted summary columns
Classical models	Logistic Regression, RBF-SVM, Random Forest, Extra Trees
Classical selection	Five-fold participant-stratified CV on development participants
CNN input	Five-second segment; 64 Mel bands; FFT 1024; hop 256
CNN optimiser	Adam, learning rate 3×10^{-4}
CNN objective	Binary cross-entropy
CNN training	Batch size 16; maximum 60 epochs; early stopping
CNN regularisation	Dropout 0.45 and 0.35; batch normalisation in first two stages
Decision threshold	0.5 for final held-out predictions

D. Reproducibility Configuration

The key settings below summarise the implementation recorded in the project notebook. They are included to make the experimental setup easier to inspect; exact software versions, random seeds, and a machine-readable participant split manifest should be supplied with any public reproduction package.

VII. Limitations and Future Work

The experiment uses one dataset and one saved participant-level train/test split, with only 17 held-out participants. The healthy-control class combines young and elderly cohorts; age and cohort composition may therefore influence the acoustic patterns. Recording devices, environments, tasks, language, and speaking styles may also affect the signal. The CNN uses only the first five seconds of each recording, potentially omitting useful information later in the utterance.

The notebook does not report external-dataset validation, prospective clinical evaluation, probability calibration, or repeated independent test splits. Future work should evaluate additional participants and independent datasets, use repeated grouped or nested grouped validation, test different segment-selection strategies, assess recording-condition robustness, and examine calibration and uncertainty. Clinical application would require prospective validation and expert oversight.

VIII. Conclusion

This project implemented two participant-aware voice-classification pipelines for the IPVS collection: a classical pipeline using 115 handcrafted acoustic feature columns and a CNN using five-second Log-Mel spectrograms. The RBF-SVM achieved 0.9558 recording-level accuracy and 0.9412 participant-level accuracy. The CNN achieved 0.9690 recording-level accuracy and the same participant-level accuracy. The models therefore show promising discrimination within the saved experiment, but the small held-out participant cohort, single dataset, and single split prevent broad generalisation claims. The work is an experimental research prototype and not a clinical diagnostic system.

Author Contributions

Suvinay Foujdar: project coordination, integration of the experimental pipeline, experimental design, and manuscript preparation. Akshat Goyal: dataset organisation, audio preprocessing, handcrafted-feature extraction, and classical machine-learning experiments. Navya Loshali: literature review, Log-Mel CNN experiments, result visualisation, and manuscript review.

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